DELIVERABLE 3: PRESENTATION

Recommendation Systems for Personalized Content Discovery

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PROBLEM OVERVIEW

The Challenge of Choice

In a library of 17,000+ titles, users face **choice paralysis**. The Netflix Prize challenged the industry to predict preferences with high accuracy.

- **Dataset:** 100M+ ratings from 480k users.
- **Goal:** Optimize content discovery through latent preference mapping.

APPROACH: DATA STRATEGY

Sparsity Check

98.82% of the interaction matrix is unobserved. Requires factorization to fill the gaps.

Thresholding

Filtered for stability: Users with ≥50 ratings and Movies with ≥500 ratings.

Sampling

Used a dense subset of 500k interactions for optimal RAM performance in Colab.

APPROACH: MODEL DESIGN

Model A: SVD

Singular Value Decomposition

Maps users and movies into a shared latent factor space. Captures hidden thematic interactions (e.g., genre affinity, director style).

Model B: Baseline

ALS Bias Model

Isolates global biases: User leniency and movie popularity. Serves as a control to measure personalization lift.

EXPERIMENTAL RESULTS

Architecture	RMSE (Accuracy)	MAP@10 (Ranking)	Complexity
SVD Factorization	0.9816	0.9523	Moderate
BaselineOnly (ALS)	**0.9744**	**0.9544**	Low

Baseline biases represent the strongest signal in user rating behavior on the platform.

RECOMMENDATION EXAMPLES

Case Study: User 345891 (A random user from the dataset)

Success Outcome: Predicted a score of 3.78 for Movie 2043 (Ground Truth: 4.0).

The model accurately placed this title at the top of the user's recommended carousel, demonstrating high ranking precision.

Top-K ranking quality is verified at 95% accuracy.

KEY INSIGHTS

- **RMSE != Ranking Precision:** Absolute error in rating prediction is decoupled from the quality of the top-item carousel ranking.
- **Bias Dominance:** Global trends and user leniency are more predictive than complex interaction vectors in sparse settings.
- **Regularization:** Shrinking extreme predictions towards the mean prevents jarring false positives, maintaining user trust.

Future Improvements & Scalability

1. Two-Tier Dual Framework: Replace the single-stage exhaustive prediction approach with a decoupled architecture.
2. Contextual & Temporal Features: Incorporate interaction timestamps and device context into the latent factor vectors to capture shifting user interest patterns over time.